50 Math Word Problems with Step-by-Step Solutions

A practical problem-solving workbook for computer science graduates

Topics included	What you will practise
Arithmetic & algebra	Percentages, ratios, averages, equations, mixtures, age and digit problems
Rates & scheduling	Work rates, speed, throughput, LCM, clock arithmetic and time calculations
Probability & counting	Basic probability, expected value, combinations, PINs and Bayes-style reasoning
Computer science applications	Binary, storage, network transfer, algorithm growth, loops, reliability and graphs

How to use this workbook

Try each question before reading its solution. Write down the information given, choose a suitable equation or counting method, calculate carefully, and check whether the answer makes sense. The questions range from foundational arithmetic to practical quantitative reasoning used in technical interviews and software work.

Note: One question intentionally illustrates how inconsistent-looking real-world inputs can produce fractional prices; the correct conclusion is that the supplied data do not yield whole-rupee prices.

Part 1 — Arithmetic, Ratios and Algebra

1. Percentage and discounts

A laptop costs ■60,000. A student gets a 12% discount and then pays 18% GST on the discounted price. What is the final price?

Solution: Discount = 12% of 60,000 = ■7,200. Discounted price = ■60,000 – ■7,200 = ■52,800. GST = 18% of 52,800 = ■9,504. Final price = 52,800 + 9,504 = ■62,304.

2. Percentage change

A software team's bug count falls from 240 bugs to 180 bugs after a release. What is the percentage decrease?

Solution: Decrease = 240 – 180 = 60. Percentage decrease = (60 ÷ 240) × 100 = 25%.

3. Ratio

A project has frontend and backend tasks in the ratio 3:5. If there are 64 tasks altogether, how many are frontend tasks and how many are backend tasks?

Solution: Total ratio parts = 3 + 5 = 8. One part = 64 ÷ 8 = 8. Frontend = 3 × 8 = 24 tasks; backend = 5 × 8 = 40 tasks.

4. Average

Five daily response-time measurements are 120, 135, 110, 145 and 140 milliseconds. What is the average response time?

Solution: Sum = 120 + 135 + 110 + 145 + 140 = 650. Average = 650 ÷ 5 = 130 milliseconds.

5. Weighted average

A candidate scores 70 in a test worth 30% and 86 in a test worth 70%. What is the weighted score?

Solution: Weighted score = 70(0.30) + 86(0.70) = 21 + 60.2 = 81.2.

6. Profit and loss

A company buys a server for ■80,000 and sells it for ■92,000. Find the profit percentage on cost.

Solution: Profit = 92,000 – 80,000 = ■12,000. Profit percentage = (12,000 ÷ 80,000) × 100 = 15%.

7. Simple interest

A student borrows ■20,000 at 6% simple interest per year for 2.5 years. How much interest is due?

Solution: Simple interest = P × R × T ÷ 100 = 20,000 × 6 × 2.5 ÷ 100 = ■3,000.

8. Compound growth

A cloud bill of ■10,000 grows by 10% each month for two months. What is the bill after two months if the increase compounds?

Solution: After month 1: 10,000 × 1.10 = ■11,000. After month 2: 11,000 × 1.10 = ■12,100.

9. Unit conversion

A data pipeline processes 2.4 GB per minute. Assuming 1 GB = 1,000 MB, how many MB does it process in 25 minutes?

Solution: 2.4 GB/min = 2,400 MB/min. In 25 minutes: 2,400 × 25 = 60,000 MB.

Part 2 — Rates, Time and Scheduling

10. Work rate

One developer can finish a task in 12 hours and another in 18 hours. If they work at constant rates together, how long will they take?

Solution: Rates are 1/12 and 1/18 task per hour. Combined rate = 1/12 + 1/18 = 5/36. Time = 1 ÷ (5/36) = 36/5 = 7.2 hours, or 7 hours 12 minutes.

11. Work with a delayed start

A tester can test a release in 8 hours. A second tester can do it in 12 hours. The first works alone for 2 hours, then both work together. How much more time is needed?

Solution: First tester completes 2/8 = 1/4. Remaining work = 3/4. Combined rate = 1/8 + 1/12 = 5/24 per hour. Additional time = (3/4) ÷ (5/24) = 18/5 = 3.6 hours.

12. Speed and distance

A delivery robot travels 18 km at 6 km/h and returns along the same route at 9 km/h. What is its average speed for the full trip?

Solution: Total distance = 36 km. Time outward = 18/6 = 3 h; return = 18/9 = 2 h. Average speed = total distance ÷ total time = 36/5 = 7.2 km/h.

13. Relative speed

Two automated carts are 300 metres apart and move toward each other at 4 m/s and 6 m/s. How long until they meet?

Solution: Closing speed = 4 + 6 = 10 m/s. Time = 300 ÷ 10 = 30 seconds.

14. Train/platform problem

A 180 m train passes a stationary sensor platform in 12 seconds. What is its speed in km/h?

Solution: Speed = 180/12 = 15 m/s. Convert to km/h: 15 × 3.6 = 54 km/h.

15. Mixture

A lab needs 20 litres of a 30% saline solution. It has a 20% solution and a 50% solution. How many litres of each should be mixed?

Solution: Let x be litres of 50% solution; 20 – x is 20% solution. Salt equation: 0.50x + 0.20(20 – x) = 0.30(20). Thus 0.30x = 2, so x = 6■ L. The 20% solution needed is 13■ L.

16. Linear equation

An online course charges a fixed registration fee plus ■250 per workshop. Four workshops cost ■1,700. Find the registration fee.

Solution: Let fixed fee be F. F + 4(250) = 1,700, so F = 1,700 – 1,000 = ■700.

17. Two unknowns

A team buys 7 keyboards and 5 mice for ■8,900. Each keyboard costs ■1,000 more than a mouse. Find each price.

Solution: Let mouse price be m; keyboard price is m + 1,000. Then 7(m + 1,000) + 5m = 8,900. So 12m = 1,900, m = ■158.33… and keyboard = ■1,158.33…. The figures imply fractional-rupee prices, so the given data do not produce whole-rupee prices.

18. Age problem

A mentor is three times as old as an intern. In 8 years, the mentor will be twice the intern's age. Find their current ages.

Solution: Let intern age = x; mentor age = 3x. In 8 years: 3x + 8 = 2(x + 8). Thus x = 8. Intern is 8 and mentor is 24 years old.

19. Consecutive integers

The sum of three consecutive integers is 72. What are they?

Solution: Let the integers be n – 1, n, n + 1. Their sum is 3n = 72, so n = 24. The integers are 23, 24 and 25.

Part 3 — Number Theory and Number Puzzles

20. HCF/GCD

Two devices send status updates every 24 seconds and 36 seconds. If they send an update together now, after how many seconds will their update times coincide again?

Solution: They coincide at multiples of both intervals. Find LCM(24, 36). Since $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$, LCM = $2^3 \times 3^2 = 72$ seconds.

21. LCM scheduling

A backup runs every 15 hours and a security scan every 20 hours. Both run at noon today. After how many hours will they run together again?

Solution: LCM(15, 20) = 60. They will coincide again after 60 hours.

22. Remainders

A batch contains fewer than 100 files. When counted in groups of 6, 2 files remain; in groups of 8, 4 remain. What is the smallest possible batch size greater than 20?

Solution: Numbers that leave remainder 2 modulo 6 are 2, 8, 14, 20, 26, 32, … Of these, 26 leaves remainder 2 modulo 8, not 4; 20 leaves remainder 4 modulo 8 but is not greater than 20. Next candidates 32 leaves remainder 0 modulo 8. Check 44: remainder 2 mod 6 and 4 mod 8. Smallest greater than 20 is 44.

23. Digit puzzle

A two-digit code has digits that sum to 11. Reversing the digits makes the number 27 larger. Find the original code.

Solution: Let tens digit be a and units digit b. a + b = 11. Reversed minus original = 9(b – a) = 27, so b – a = 3. Solving gives b = 7 and a = 4. Original code = 47.

24. Prime factors

A server assigns an integer ID of 360 to a test job. Express 360 as a product of prime factors.

Solution: $360 = 36 \times 10 = (2^2 \times 3^2) \times (2 \times 5) = 2^3 \times 3^2 \times 5$.

Part 4 — Probability and Counting

25. Probability

A box contains 5 red, 3 blue and 2 green test tokens. One token is chosen uniformly at random. What is the probability it is not blue?

Solution: Total tokens = 10. Non-blue tokens = 5 + 2 = 7. Probability = 7/10 = 0.7 = 70%.

26. Two-event probability

A fair six-sided die is rolled twice. What is the probability the total is 7?

Solution: There are 36 equally likely ordered outcomes. The six outcomes summing to 7 are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). Probability = 6/36 = 1/6.

27. Without replacement

A bag contains 4 faulty and 6 working USB drives. Two are selected without replacement. What is the probability both are faulty?

Solution: Probability = (4/10) × (3/9) = 12/90 = 2/15.

28. Expected value

A monitoring alert costs ■100 to investigate. There is a 5% chance an alert represents a real incident worth preventing a ■5,000 loss. If investigating prevents the loss whenever the incident is real, what is the expected net value of investigating?

Solution: Expected avoided loss = 0.05 × ■5,000 = ■250. Subtract investigation cost ■100. Expected net value = ■150 per alert, under the stated assumptions.

29. Combinations

A team of 3 students must be chosen from 8 candidates. How many different teams are possible if order does not matter?

Solution: Use combinations: C(8,3) = 8×7×6 ÷ (3×2×1) = 56 teams.

30. Password counting

A PIN has 4 digits, each from 0 to 9, and digits may repeat. How many PINs are possible, including those beginning with zero?

Solution: Each of the four positions has 10 choices. Total = 10■ = 10,000 PINs.

31. No repeated digits

A 3-digit access code uses digits 1 through 9 with no repeated digit. How many codes are possible?

Solution: First position: 9 choices; second: 8; third: 7. Total = 9×8×7 = 504.

Part 5 — Computer Science Quantitative Problems

32. Set overlap

Of 100 applicants, 60 know Python, 45 know SQL, and 25 know both. How many know at least one of the two?

Solution: Use inclusion-exclusion: $|P \cup S| = |P| + |S| - |P \cap S| = 60 + 45 - 25 = 80$ applicants.

33. Venn diagram reasoning

In a class of 50, 28 study AI, 22 study cybersecurity and 10 study both. How many study neither subject?

Solution: At least one = 28 + 22 – 10 = 40. Neither = 50 – 40 = 10 students.

34. Binary numbers

A sensor stores the decimal value 45. Convert 45 to binary.

Solution: 45 = 32 + 8 + 4 + 1 = 2■ + 2^3 + 2^2 + 2■. Therefore the binary representation is 101101.

35. Binary arithmetic

A system adds the binary numbers 1011■ and 0110■. What is the result in binary and decimal?

Solution: 1011■ = 11 and 0110■ = 6. Sum = 17 decimal = 10001■.

36. Bit storage

A log record uses 12 bits per event. How many bytes are needed to store 1,000 events if records are packed continuously and 1 byte = 8 bits?

Solution: Total bits = 12 × 1,000 = 12,000 bits. Bytes = 12,000 ÷ 8 = 1,500 bytes.

37. Algorithm growth

Algorithm A performs $3n^2$ operations and Algorithm B performs 100n operations. At n = 20, which performs fewer operations?

Solution: A: 3×20^2 = 1,200. B: 100×20 = 2,000. Algorithm A performs fewer operations at n = 20.

38. Crossover point

Algorithm A takes n^2 operations and Algorithm B takes $20n$ operations. For what positive n are their operation counts equal, and which is smaller beyond that point?

Solution: Set $n^2 = 20n$. For positive n, divide by n to get $n = 20$. For $n > 20$, $20n < n^2$, so Algorithm B has fewer operations.

39. Loop counting

A program runs an inner loop of 5 iterations for every iteration of an outer loop that runs n times. How many inner-loop executions occur when $n = 120$?

Solution: The inner loop runs 5 times per outer iteration, so total = $5n = 5 \times 120 = 600$ executions.

40. Nested loop

A program compares every pair of distinct items in an array of 10 items, counting each unordered pair once. How many comparisons occur?

Solution: Number of unordered pairs = $C(10,2) = 10 \times 9/2 = 45$ comparisons.

41. Queue throughput

A server processes 240 requests per minute. At a steady rate, how long does it take to process 3,000 requests?

Solution: Rate = 240/60 = 4 requests per second. Time = 3,000 ÷ 4 = 750 seconds = 12 minutes 30 seconds.

42. Latency and throughput

A service completes 80 requests per second. Assuming constant throughput, how many requests can it complete in 7.5 minutes?

Solution: 7.5 minutes = 450 seconds. Requests = $80 \times 450 = 36{,}000$.

43. Database storage

Each database row uses 1.5 KB on average. Approximately how much storage do 80,000 rows require, using decimal units?

Solution: Storage = $1.5 \times 80{,}000 = 120{,}000$ KB. Since 1 MB = 1,000 KB, this is 120 MB, excluding indexes and overhead.

44. Network transfer

A 900 MB file is transferred over a 30 MB/s connection. Ignoring overhead, how long will the transfer take?

Solution: Time = data size ÷ rate = 900 ÷ 30 = 30 seconds.

45. Reliability

A service has a 99.5% chance of working correctly on any one independent request. What is the probability that all 10 requests work correctly?

Solution: Probability = $0.995^{10} \approx 0.9511$. So the chance is approximately 95.11%, assuming independence.

Part 6 — Applied Reasoning and Optimization

46. Complement probability

A test catches a bug with probability 0.8. If three independent tests are run, what is the probability at least one catches it?

Solution: Probability one test misses it = 0.2. All three miss = $0.2^3 = 0.008$. At least one catches it = $1 - 0.008 = 0.992 = 99.2\%$.

47. Bayes-style diagnostic

A bug occurs in 2% of builds. A detector flags 90% of buggy builds and falsely flags 5% of clean builds. Among flagged builds, what fraction are expected to be genuinely buggy?

Solution: For 1,000 builds: buggy = 20, of which 18 are flagged. Clean = 980, of which 49 are falsely flagged. Total flagged = 18 + 49 = 67. Fraction truly buggy = 18/67 ≈ 26.9%.

48. Graph degree

In a network graph, 6 computers have degrees 2, 2, 3, 3, 4 and x. The sum of degrees must be even. What values of x parity are possible?

Solution: Known degree sum = 2 + 2 + 3 + 3 + 4 = 14, which is even. Therefore x must be even so the total remains even. This is necessary; a specific graph may impose additional constraints.

49. Shortest path arithmetic

A robot route has edges of 4 m, 7 m and 5 m. Another route has edges of 8 m and 6 m. Which route is shorter, and by how much?

Solution: Route 1 = 4 + 7 + 5 = 16 m. Route 2 = 8 + 6 = 14 m. Route 2 is shorter by 2 m.

50. Coordinate distance

A robot moves from grid point (2, 3) to (8, 11). What is the straight-line distance?

Solution: Distance = $\sqrt{((8–2)^2 + (11–3)^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$ units.

51. Linear scaling

A model training job takes 18 minutes for 6,000 records. If runtime scales linearly, how long will 15,000 records take?

Solution: Scale factor = 15,000/6,000 = 2.5. Runtime = 18 × 2.5 = 45 minutes.

52. Resource allocation

A cloud budget is ■48,000. Each small instance costs ■2,000 and each large instance costs ■6,000. If the team needs 12 instances and spends the entire budget, how many of each should it rent?

Solution: Let s + l = 12 and 2,000s + 6,000l = 48,000. Divide cost equation by 2,000: s + 3l = 24. Subtract s + l = 12 to get 2l = 12, so l = 6 and s = 6.

53. Optimization

A rectangular testing area has 40 m of fencing for its four sides. What dimensions give the largest area?

Solution: Let length = l and width = w. 2l + 2w = 40, so l + w = 20. Area is maximized when the rectangle is a square: l = w = 10 m. Maximum area = 100 m².

54. Work backwards

After spending one-third of a project budget and then ■4,000 more, ■8,000 remains. What was the original budget?

Solution: Let budget be B. Remaining: B – B/3 – 4,000 = 8,000. Thus 2B/3 = 12,000, so B = ■18,000.

55. Clock arithmetic

A scheduled job starts at 22:45 and runs for 3 hours 50 minutes. At what time does it finish the next day?

Solution: 22:45 + 1 hour 15 minutes = 00:00; remaining duration = 2 hours 35 minutes. Finish time is 02:35 the next day.

56. Probability from a dataset

A dataset has 500 records, of which 35 are mislabeled. If one record is sampled uniformly, what is the probability it is correctly labeled?

Solution: Correct records = 500 – 35 = 465. Probability = 465/500 = 0.93 = 93%.

57. Median and outliers

Five test runtimes in seconds are 4, 5, 5, 6 and 30. Find the median and explain why it may be more representative than the mean.

Solution: Sorted values are already 4, 5, 5, 6, 30. Median = 5 seconds. Mean = 50/5 = 10 seconds. The unusually high 30-second value pulls the mean upward, while the median is less affected by an outlier.

58. Simple optimization

A team has 100 engineer-hours. Feature A needs 12 hours and Feature B needs 8 hours. If they can deliver only whole features, what is the maximum number of features if they choose only Feature B?

Solution: Each B feature uses 8 hours. Maximum = floor(100/8) = 12 features, using 96 hours with 4 hours left.

59. Pigeonhole principle

A system logs events with one of 7 weekday labels. What is the minimum number of events needed to guarantee at least 5 events share the same weekday label?

Solution: To avoid having 5 in any label, each of 7 labels can have at most 4 events: 7×4 = 28. One more event guarantees a label has at least 5. Answer: 29 events.

End of workbook — keep practising by changing the numbers and solving each problem again.